

RAJIV RANJAN JHA

AI Engineer · Agentic Systems, RAG & LLMOps

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AI Engineer, 4 yrs — production **agentic systems, RAG, LLMOps**. Multi-agent RAG at **<800ms TTFT, >95% faithfulness, 85%+ recall@5**. **LoRA/QLoRA** fine-tuning on a **single 8GB GPU** — **2× throughput**, spend **under \$0.50**. Owns the full path: data → training → eval → serving → infra.

SKILLS

Agents LangGraph · LangChain · CrewAI · Google ADK · MCP · custom ReAct loops · function/tool calling · Pydantic structured outputs · prompt engineering

RAG & Retrieval Hybrid dense + BM25 · cross-encoder re-ranking · GraphRAG · Schema-RAG · FAISS · pgvector · Pinecone · Sentence-Transformers · recall@k tuning

Eval & Safety Faithfulness/groundedness · hallucination rate · ROUGE · BERTScore · exact match · evidence gates · golden sets · PII redaction · input/output guardrails

Serving & Obs vLLM · llama.cpp/GGUF · Ollama · FastAPI · SSE streaming · Langfuse · LangSmith · Helicone · MLflow

Fine-tuning LoRA/QLoRA (PEFT) · rsLoRA · Unsloth · BF16 · gradient checkpointing · CPT/SFT · Hugging Face · PyTorch

ML LightGBM · time-series forecasting (WAPE) · YOLO · ArcFace · diffusion · statistical modelling

Data & Cloud BigQuery · PostgreSQL/AlloyDB · Medallion (Bronze/Silver/Gold) · Dataform · ETL/ELT · MinHash dedup · GCP / Google Cloud (Cloud Run, Vertex AI) · Azure/Databricks · Docker · CI/CD · cloud-agnostic deploys

EXPERIENCE

Senior Data Scientist / AI Engineer · Alkye Technologies

Sep 2024 – Present

Merchmix — AI inventory-planning SaaS

- **70–80% of analyst queries eliminated** — multi-agent, multi-modal RAG chatbot: **LangGraph** orchestration + **CrewAI** role delegation over Google ADK and MCP tools.
- **85%+ recall@5, >95% faithfulness, <800ms TTFT** — **hybrid retrieval**: dense vectors (**pgvector**) + **BM25, cross-encoder re-rank**, schema-aware over BigQuery/PostgreSQL.
- **Langfuse + LangSmith** tracing on every agent run — per-run token spend, latency, tool-call traces.
- **60% dashboard query cost cut** — Medallion (Bronze/Silver/Gold) pipeline, BigQuery → AlloyDB via API ingestion, zero freshness loss.
- **86% WAPE** forecast accuracy — **LightGBM** time-series across brand/category hierarchies, daily + weekly SKU demand.
- Latency cut under load — Django **sync** → **async I/O**, **Redis** caching, load-balanced **Docker/Cloud Run** via CI/CD.

Fittora — AI made-to-order tailoring platform (Australian client)

- **95%+ identity-retrieval accuracy** — **ArcFace** cosine similarity, matching published **SOTA 94–99%**.
- Solved **multi-angle identity drift** — custom annotation strategy + targeted model fine-tuning on the failure cases.
- **<1s mockup generation, 3–5× faster** than the baseline **diffusion** model — dual-path architecture: primary face-swap with model fallback.
- **99%+ generation success rate** held in production — fallback path absorbs primary-model failures.

Data Scientist · SK Associates

Jul 2022 – Sep 2024

- Shipped custom **YOLO** object-detection models to production for client computer-vision applications — owned the full path end to end.
- Built the **annotation and training pipeline** — dataset curation, labelling standards, iteration on failure cases.
- Owned **model serving and deployment** for CV workloads, from trained weights to a running client-facing service.
- Built the **SQL and ETL pipelines** feeding reporting — replaced raw-table extracts with modelled, query-ready data.
- Replaced raw-table reporting with **executive dashboards** surfacing revenue and profit drivers for banking-client leadership decisions.

PROJECTS

LLMOps Fine-Tuning Pipeline — e-commerce domain adaptation *PyTorch · Unsloth · LoRA/PEFT · vLLM · llama.cpp · MLflow · LangChain · FastAPI*

- **TinyLlama 1.1B** domain-adapted on a **single 8GB consumer GPU** — **LoRA rank-32 (rsLoRA)**, **2× throughput** via Unsloth BF16 + gradient checkpointing.
- Served through **vLLM** and **llama.cpp/GGUF**; runs and checkpoints tracked in **MLflow** model registry.
- **ROUGE** and **BERTScore** gains vs base, plus **exact-match** scoring on a held-out golden set via **LangChain** eval chains.
- **14K → 6.4K docs (54% rejected)** — 7-filter **guardrail** and curation engine: **PII redaction**, MinHash dedup, quality scoring, percentile pruning, full audit trails.
- Production FastAPI service — job management, **SSE log streaming**, web UI; domain-portable by YAML with **zero code changes**.

Autonomous Coding Agent — evidence-gated execution harness *Asyncio · Vertex AI (Gemini) · Playwright · Pydantic · Helicone · 324 tests*

- **15 typed tools** in a custom **ReAct loop** (plan → act → observe → verify) — **function calling** with **Pydantic structured outputs**, no LLM awareness of the tool layer.
- **Caught false success in 3 of 9 (33%)** apparently-passing runs — **evidence gates** independently re-execute success predicates; verification via a real browser's accessibility tree.
- **94% tool-output compression** — observation compiler preserving error spans, holding context at **22K of a 1M window**; total spend **under \$0.50**.
- **324 tests**; **Helicone** for per-call cost and latency tracking.

EDUCATION

B.Tech · CV Raman Global University, Bhubaneswar

Jul 2018 – Jun 2022